

Rohit Agarwal

Full Time Ph.D. Research Scholar

ra.22cs1102@phd.nitdgp.ac.in | +91-9775193530

National Institute of Technology Durgapur, West Bengal - 713209

 LinkedIn,  Google Scholar

RESEARCH INTEREST

• Biomedical Image Processing • Computer Vision • Multi-modal AI • Generative AI • Explainable AI

EXPERIENCE

Teaching Assistance | 1st September, 2022 - Present
National Institute of Technology, Durgapur

Assistant Teacher (Computer) | 4th April, 2022 - 31st August, 2022
Shemford Futuristic School, Burdwan

EDUCATION

National Institute of Technology | Ph.D. in Computer Science and Engineering
August, 2022 - Present | Durgapur
Design of Machine Learning-based CAD System for Medical Image Analysis

University Institute of Technology | M.E. in Computer Science and Engineering
2021 | Burdwan
TCPA : 87.28

Academy of Technology | B.Tech in Computer Science and Engineering
2019 | Kolkata
DGPA : 7.98

East West Model School
ISC 2015 | Burdwan
85.83%

East West Model School
ICSE 2013 | Burdwan
88.33%

SKILLS

PROGRAMMING

Python • Java • C • C++ • MySQL • HTML • CSS • Javascript

SOFTWARE TOOLS

PyTorch • Tensorflow • Keras • Numpy • $\LaTeX$

PUBLICATIONS

JOURNALS

- **Deep quasi-recurrent self-attention with dual encoder-decoder in biomedical CT image segmentation**

Agarwal, R., Chowdhury, A., Chatterjee, R. K., Chel, H., Murmu, C., Murmu, N., Nandi, D. (2024)
IEEE Journal of Biomedical and Health Informatics, 28(12), 7195–7205.
<https://doi.org/10.1109/jbhi.2024.3447689>

- **Multi-scale dual-channel feature embedding decoder for biomedical image segmentation**

Agarwal, R., Ghosal, P., Sadhu, A. K., Murmu, N., Nandi, D. (2024)
Computer Methods and Programs in Biomedicine, 257(108464), 108464.
<https://doi.org/10.1016/j.cmpb.2024.108464>

- **Compound attention embedded dual channel encoder-decoder for ms lesion segmentation from brain MRI**

Ghosal, P., Roy, A., Agarwal, R., Purkayastha, K., Sharma, A. L., Kumar, A. (2024)

Multimedia Tools and Applications..

<https://doi.org/10.1007/s11042-024-20416-3>

CONFERENCES

- **Towards Robust Category-level Articulation Pose Estimation via Integrated Differentiable Rendering**

Yu, X., Jiang, H., Zhang, L., Huo, Y., Wu, L., Wei, Y., Agarwal, R., Shende, H., Liu, L., Ou, L.

2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2025), Hyderabad, India.

(Accepted)

- **Generalizable Articulated object perception with superpoints**

Yu, Q., Hao, C., Yuan, X., Zhang, L., Liu, L., Agarwal, R., Huo, Y., Lu, C..

2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2025), Hyderabad, India.

(Accepted)

- **Deep Entropy-Sampled O-Net for Joint Retinal Cup-Disc Segmentation in Glaucoma Detection**

Chowdhury, A., Agarwal, R., Das, A., Jalan, A., Biswas, S., Nandi, D.

4th International Conference on Intelligent Systems and Sustainable Computing, Hyderabad, India. (Accepted)

- **Spiking Neural Network: Techniques, Tools and Trends**

Agarwal, R., Ghosal, P., Murmu, N. & Nandi, D.

Fourth International Conference on Advanced Computational and Communication Paradigms (ICACCP - 23),

Sikkim, India.

https://doi.org/10.1007/978-981-99-4284-8_16

- **Automatic detection of COVID-19 from Chest X-ray images using Deep Learning model**

Das, A., Agarwal, R., Singh, R. Chowdhuri, A. & Nandi, D.(17-18 June, 2021)

International Conference on Computational Intelligence and Computing Applications (ICCIICA - 21), Nagpur, India.

<https://doi.org/10.1063/5.0076882>

- **U-Net Based Optic Cup and Disc Segmentation From Retinal Fundus Images via Entropy Sampling**

Chowdhuri, A., Agarwal, R., Das, A., & Nandi, D.(22-24 March, 2021)

Third International Conference on Advanced Computational and Communication Paradigms (ICACCP - 21), Sikkim, India.

https://doi.org/10.1007/978-981-16-4369-9_47

AWARDS & ACHIEVEMENTS

- Reviewer for Various International Conferences and Journals
- Invited Distinguished Speaker for Various International Conferences and Workshops
- Qualified WBSET 2022
- Qualified UGC-NET 2022
- Participated in the Exhibition of "4th Regional Science & Technology Congress (Western Region), 2019" organized jointly by The University of Burdwan and Department of Science & Technology and Biotechnology, Government of West Bengal
- Qualified GATE 2019 with an AIR of 5054 and GATE Score 492
- Completed Python Training offered by the Spoken Tutorial Project, IIT Bombay, funded by National Mission on Education through ICT, MHRD, Govt., of India
- Runners' Up in Inter-College Football Tournament
- Completed Deep Learning with Tensorflow provided by CognitiveClass